



How attitude, shared tacit assumptions, espoused values, subjective norms, behavioural intention and knowledge influence information security culture

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Abstract

Threat vectors to information assets of firms continue to multiply especially with rapid advances in ICT capabilities. Financial services firms are one of the prime targets of information security breaches because of the increased probability of the miscreants realising pecuniary benefits. The cultivation of information security culture is viewed as the most effective safeguard of warding off information security breaches. This study examined the impact that knowledge, attitude, shared tacit assumptions, subjective norms, espoused values and behavioural intention exert on financial services information security culture. The study's participants were drawn from four financial services firms in Zimbabwe. The questions that constituted the questionnaire that was sent to the respondents were derived from the extant literature on information security culture. The regression analysis performed on the responses to the questionnaire showed that knowledge, attitude, subjective norms, shared tacit assumptions and behavioural intention are all significant predictors of financial services information security culture. However, the study established that espoused values are not a significant predictor of financial services information security culture. The study suggested that financial services should leverage the constructs that the study established to be significant predictors of financial services information security culture especially improving their employees' information security knowledge and attitudes which in turn influence the other constructs.

Keywords Information security culture · Attitude shared tacit assumptions · STA espoused values · EV subjective norms · SN behavioural intention · BI knowledge · Financial services

1 Introduction

Information and communication technology (ICT) capabilities have improved exponentially over the past quarter of a century mainly driven by artificial intelligence and machine learning. These advances in ICT capabilities have also brought to the fore the unintended consequences of having

the same ICT capabilities being leveraged by miscreants to cause information security breaches on firms. The financial services sector is one of the prime targets of information security breaches since it hoards liberal quantities of personally identifiable information of its clients and the probability of malefactors reaping massive pecuniary gains is huge.

Coincidentally, information security culture which is forecast to be the most effective safeguard against information security breaches has been on the agenda and growing since about the last 25 years. Borrowing from the domains of Psychology and Business Management particularly from Schein's Model [1], advanced knowledge, shared tacit assumptions, espoused values and behavioural artefacts as determinants of information security culture. Later on [2], proposed attitude, subjective norms and behavioural intention as other predictors of information security culture. The challenge with these constructs is that they have generally

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been proposed and discussed in anecdotal studies with very limited empirical verification. Specific to the Zimbabwean financial services context, only the study of [3] which tested just knowledge of all the seven constructs is the only related empirical study we are aware of.

To empirically examine whether or not knowledge, attitude, subjective norms, shared tacit assumptions, espoused values and behavioural intention are predictors of information security culture (represented by behavioural artefacts) in the financial services sector, we recruited 386 respondents from four financial services firms in Zimbabwe using the purposeful sampling methodology. Questionnaires composed of questions derived from extant literature on the constructs of this study were electronically conveyed to 400 participants with their responses captured on ODK. 2.0. The study assumed that the selected financial services sector employees to have adequate knowledge of its constructs, which is not expected of laymen. This study was commissioned to cover the gap of empirical information security studies, especially in the financial services sector context in Zimbabwe.

2 Literature review

2.1 Attitude

Strong attitudes formed from deep calculations are usually predictable antecedents of behaviour [4]. Such strong attitudes are acquired through either interacting with peers or poring over their conduct [4]. The social repercussions flowing from the deeds in question will determine the extent of the attitude-behaviour link [4]. The relationship between attitude and subjective norms herein referenced is in line with tenets of the Theory of Reasoned Action (TRA) which postulates that the two are significant predictors of the behavioural intention to engage in behaviour [5]. Attitudes require activation in a person and the frequency of the expression of an attitude the more robust it becomes [6].

What makes attitude an important construct within this study is its influence on behavioural intention [4] and behaviour [7]. This is so because the thrust of information security culture is the transformation of employees' behaviour especially through information security awareness [6].

2.2 Shared Tacit assumptions

Shared tacit assumptions (STA) are the indicator of culture that provides a scheme for comprehending and expose human encounters [8]; [2]. Tacit assumptions assist to construct and recognise culture, and where they are common to groups, can convey them closer in accomplishing a familiar

task [8]; [2]. They are established through the interfacing of the firm and its environs, are held to be reasonable and factual, are disseminated and coached to peers, are taken for a ride, and are hard to alter and epitomise the most profound level of culture [9].

Shared tacit assumptions are an intricate indicator of information security culture that forms its heart is "the set of assumptions held in common and taken for granted in an organisation" [10]. Van Niekerk and von Solms contend that the shared tacit assumptions can only be ingrained in an organisation through the establishment of information security policy by the firms' executives. The shared tacit assumptions can be strengthened through the education of employees [11]; [2].

2.3 Espoused values

Espoused values (EV) play a vital role in providing the operational campus of the organisation as well as the anticipated code of conduct of its employees [12]. Of late, espoused values have been tied to firm outputs, for instance, better financial results [13], social cohesion in multi-national companies' subsidiaries [14], and employees' dedication to firms should they deem fit between the latter's espoused and enacted morals [15]. Given the communal nature of espoused values, top management is inclined to advocate for the ones attuned to existing social norms, thereby making them a perfect portrayal of the firm's mores [16]; [15]. In accordance with the value-attitude-behaviour theory, though values exert influence on behaviour, that influence is intermediated by attitude [17].

Despite how espoused values are fashioned, their existence does influence how stakeholders perceive the firm [12]. Any variance between the espoused values and what transpires on the ground are laid bare leaving stakeholders including employees to choose the ones that they deem corresponding to their own values [14].

2.4 Subjective norms

Subjective norms (SN) are structured through the garnering of beliefs of an individual's significant others (persons and cliques) [18], typically management, information technology staffers or contemporaries in an information security setting [20]. Subjective norms are context-specific [5] with environmental prompts implying the proper conduct to engage in [18]. From an information security perspective, employees strive to safeguard themselves and their property [19]. The manner in which they invoke their protection norm is however dictated upon by their atmosphere, information systems quality [18]. Should the environment signal that the technology being used by the employee are

of superior quality, “users will intend to engage in secure behaviour in order to meet the implicitly agreed upon shared expectations” [18:26]. In the paradox, hints of inferior systems quality will trigger reduced security norms of the users [18].

Subjective norms are a significant predictor of employees’ intentions to conform to the firm’s information security policies [20]; [17]. Subjective norms are a predictor of intention which in turn influences experienced behaviour [17]; [5]. Technology quality induces subjective norms attuned to worthwhile conduct and sways the manner in which beliefs are operationalised [18].

2.5 Behavioural intention

Behavioural intention is a portrayal of a person’s preparedness to engage in a certain behaviour [22] and there exist abundant evidence to support the intention-behaviour correlation notably in circumstances wherein the time lag between the implied behavioural intention and actual behaviour performance is limited [5].

“The stronger the intention in the direction of the actual behaviour, the more likely the behaviour will actually be carried out” [23:62]. Though extant literature corroborates the correlation between behavioural intention and actual behaviour, there exists no assurance that staff members will perform indicated intended behaviour [23].

2.6 Knowledge

Since knowledge-aligned concepts are the chief determinants of human behavioural intents, they should be included in information security risk management strategies [11]; [24]. While 50% of information security infringements occasioned by employees are traced to their deficiency in knowledge and acts of non-compliance [25], employees’ information security attitude and intention to comply with information security policies are influenced by their information security knowledge [26]; [27].

Employees require knowledge on how to safely operate devices at their disposal at the workplace as well as the attendant risks of not doing so [3]. Knowledge is a significant predictor of the successful implementation of organisational innovation while it also distinguishes employees’ attitudes towards the pursuit of information security culture [28]. Whilst knowledge is an antecedent of awareness, the effecting of information security policies and knowledge sharing by financial services firms is enabled by the presence of knowledgeable employees [29]; [30]. Providentially, financial service firms engage employees on the basis of their having baseline knowledge in peculiar fields [3].

Financial services sector employees work with confidential and delicate information and as such should be equipped with superior information security knowledge [31]. The authors advise that to improve the information security knowledge levels in the banking industry, management and staff should concentrate on information security awareness since the miscreants’ attack methods are ever evolving. Since the attacks at the firm’s information systems are usually launched by breaching corporate networks by the virtue of human conduct, staff members can transform their conduct to rein in the hazards [31].

3 Methodology

3.1 Sample and sampling method

Purposeful sampling method was used to recruit study participants. Gatekeepers were identified in the four case financial services firms who in turn identified of their colleagues who fit into the requirements for the participant for this study namely having worked for departments wherein they would interact with personally identifiable information (PII) of the financial services firms’ clients and having worked for the financial services firm for at least for at least years. Having ascertained that the financial services sector in Zimbabwe had a total headcount of 4,000 employees, the sample of 370 was arrived at used the following formula:

$$n = p\% \times q\% \times \left\{ \frac{Z}{e\%} \right\}^2$$

The quantitative sample is even above the minimum recommended for probability sampling. Saunderson et al. recommend that the threshold sample for probability sampling be 357 at 95% confidence level and 5% margin of error for a population of 5,000. Non-probability samples are inherently smaller than their probability counterpart since the former strives to make generalisations to the theory as opposed to a population with the latter [32]; [34]. Additionally, some published information security studies performed before this study successfully employed even smaller samples. Examples of such studies are [22], 127; [6], 115; [23], 136.

The questionnaire for the study was loaded onto ODK 2.0 and were emailed to the research participants to complete electronically. In total, 386 responses were obtained from the 400 selected participants from the four financial services firms in Zimbabwe.

3.2 Instrument design

The survey questions were derived from the review of extant literature on knowledge, attitude, SN, STA, EV and BI's effect on financial services information security culture. A pilot survey was performed with six respondents from the financial services firms in Zimbabwe resulting in marginal changes to the original questionnaire. The final questionnaire was loaded onto ODK 2.0 with the respondents accessing it from that platform and their responses captured thereon. Meanwhile, each question was scored using a five-point Likert scale (5-strongly agree, 1 strongly disagree). The study proposal submission form was submitted to and approved by the ethics committee of the first author's university in November 2023. All the survey responses were kept anonymous, and their participation in this study was absolutely volitional.

3.3 Data analysis

IBM Statistics Package for Social Science (SPSS) was used to analyse the data results ($N=386$). Cronbach alpha was to check for internal validity. Normality testing, correlations and multivariate regressions were adopted to analyse the variables.

4 Findings

The tests performed on the data are Reliability, Confirmatory Factor Analysis, Correlation and Multiple Regression Analysis.

4.1 Reliability

The reliability check for this study was performed using the Cronbach alpha by which measure an alpha of less than 0.6 is not satisfactory, the one above 0.8 is regarded as good while an alpha between 0.6 and 0.8 is acceptable. The alpha coefficient for the six items is 0.791 suggesting that the items have relatively high internal consistency.

4.2 Confirmatory factor analysis (CFA)

Confirmatory Factor Analysis was used to find out if observed variables contribute to latent or unobserved variables. This is the first step in factor analysis. Data with limited or no correlation between the variables are not appropriate for factor analysis. We used two criteria to test if the data are suitable for factor analysis namely Bartlett and The Kaiser-Meyer-Olkin (KMO) tests.

The Bartlett and The Kaiser-Meyer-Olkin test evaluate all available data together. The Bartlett test statistic for this study is large and significant (significance level close to 0) as desired. The KMO measure is larger than 0.5 and thus acceptable. This implies that all variables can be used in the analysis.

4.3 Correlation matrix

A Correlation Matrix is simply a rectangular array of numbers which gives that correlation coefficients between a single variable and every other variable in the investigation. The correlation coefficient between a variable and itself is always 1, hence the principal diagonal of the correlation matrix contains 1s. The correlation coefficient can range in value from -1 to 1 . The larger the absolute value of the coefficient, the stronger the relationship between the variables. The correlation coefficient for this study range between -0.033 and 0.6 as portrayed in Table 1 below.

4.4 Multiple regression analysis

Multiple Regression Analysis is used when we want to describe how much of the variance in the dependent variable can be explained by the independent variables. Only one value is important in the interpretation: Sig value. Based on the significant value, the null hypothesis (there is no relationship between two variables) is accepted or rejected. If Sig is less than 0.05, the null hypothesis is rejected. Otherwise, it is not rejected. If a null hypothesis is rejected, it means there is an impact. However, if a null hypothesis is not rejected, it means there is not an impact. The results of the Multiple Regression Analysis are shown in Table 2 below.

4.5 Significant correlations

A correlation analysis was performed to statistically connect the known variables and the unknown ones. The correlation analysis established that attitude and behavioural intention have positive moderate relationships with financial services information security culture, knowledge, shared tacit assumptions and espoused values have positive weak relationships with financial services information security culture while subjective norms have a negative moderate relationship with financial services information security culture.

4.6 Significant predictors

A multivariate regression model was run to assess whether knowledge, attitude, subject norms, shared tacit assumptions,

Table 1 Correlations

		Correlations						
		ATT	EV	BA	STA	SB	BI	K
ATT	Pearson Correlation	1	.377**	.305**	.257**	.511**	.494**	.586**
	Sig. (2-tailed)		.000	.000	.000	.000	.000	.000
	N	336	336	336	336	336	336	336
EV	Pearson Correlation	.377**	1	.271**	.535**	.364**	.479**	.274**
	Sig. (2-tailed)	.000		.000	.000	.000	.000	.000
	N	336	336	336	336	336	336	336
BA	Pearson Correlation	.305**	.271**	1	.201**	-.033	.322**	.271**
	Sig. (2-tailed)	.000	.000		.000	.553	.000	.000
	N	336	336	336	336	336	336	336
STA	Pearson Correlation	.257**	.535**	.201**	1	.382**	.372**	.126*
	Sig. (2-tailed)	.000	.000	.000		.000	.000	.021
	N	336	336	336	336	336	336	336
SB	Pearson Correlation	.511**	.364**	-.033	.382**	1	.409**	.389**
	Sig. (2-tailed)	.000	.000	.553	.000		.000	.000
	N	336	336	336	336	336	336	336
BI	Pearson Correlation	.494**	.479**	.322**	.372**	.409**	1	.364**
	Sig. (2-tailed)	.000	.000	.000	.000	.000		.000
	N	336	336	336	336	336	336	336
K	Pearson Correlation	.586**	.274**	.271**	.126*	.389**	.364**	1
	Sig. (2-tailed)	.000	.000	.000	.021	.000	.000	
	N	336	336	336	336	336	336	336

espoused values and behavioural intention were significant predictors of information security culture in the financial services sector. With the dearth of empirical studies on the effect of the constructs in influencing information security culture in the financial services sector, extant literature overwhelmingly views all the above constructs as significant

predictors of information security culture. The regression analysis established that attitude, behavioural intention, knowledge, subjective norms and shared tacit assumptions are all significant predictors of with financial services information security culture. In the paradox, espoused values were established to be an insignificant predictor of financial

Table 2 Multiple regression analysis

Coefficients ^a					
Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.
	B	Std. Error	Beta		
1 (Constant)	1,571	,327		4,799	,000
ATT	,304	,088	,230	3,449	,001
STA	,148	,064	,136	2,307	,022
EV	,134	,075	,110	1,787	,075
SB	-,467	,071	-,391	-6,585	,000
BI	,275	,081	,205	3,403	,001
K	,199	,072	,166	2,775	,006

a. Dependent Variable: BA

services information security culture. The findings are generally in line with the views of extant literature on information security culture with the exception of the one in relation to subjective norms which [35] and [36] contend to be an unpredictable construct. The result with relation to espoused values is also an exception since extant literature view them as a significant predictor of information security culture.

5 Summary of results

The current survey as one of the few seminal empirical studies that assessed the impact of knowledge, attitude, subjective norms, shared tacit assumptions, espoused values and behavioural intention of financial services information security culture. We used quantitative data from a sample of 386 respondents from four cases from the financial services sector in Zimbabwe. A purposeful sample was extracted since the constructs of information security culture are not expected to be known to laymen. The results of the survey revealed that knowledge, attitude, subjective norms, shared tacit assumptions and behavioural intention are all significant predictors of financial services information security

culture. However, contrary to the general prescripts of extant literature on financial information security culture, the survey established that espoused values are not a significant predictor of financial services information security culture.

6 Discussion and implications

6.1 Discussion

This study investigated the impact that knowledge, espoused values, shared tacit assumptions, attitude, subjective norms, and behavioural intention have on financial services information security culture (represented by behavioural artefacts). Whereas the first three constructs and behavioural artefacts have been on the information security agenda since [1], the latter three have only found their way into the information security domain less than a decade ago with [2]. Despite the difference in the timing of the constructs being recognised in the domain, the common denominator is that they have not been subjected to empirical verification in the financial services sector in Zimbabwe which has been accomplished by this study.

The study confirmed knowledge as a significant predictor of information security culture in the financial services sector. This is in line with the views of extant literature on information security culture which generally converges on the efficacy of knowledge as a predictor of information security culture. For instance [11], argues that knowledge is an antecedent of behavioural intention and as such should be included in information security strategies. Safa and von Solms posit that information security awareness, information security compliance and information security knowledge sharing are only made possible by the availability of knowledgeable employees. Providentially, financial services firms engage employees on the basis of them having baseline knowledge which is normally some tertiary education. Therefore, it comes as no surprise that knowledge was established to be a significant predictor of information security culture in the financial services sector by this study.

The study established that attitude is a significant predictor of financial services information security culture. The finding is in line with the prescripts of information security culture extant literature which concurs on the fact that attitude is a significant predictor of behavioural intention and behaviour artefacts [4]; [7]; [5]. It is also in conformity with the Value-Attitude-Behaviour Theory which posits that values impact on behaviour through the intermediation of attitude [17]. Similarly, the study found that shared tacit assumptions are a significant predictor of financial services information security culture. Behavioural intention was also established to be a significant predictor of financial services information security culture. The finding chimes well with the Theory of Reasoned Action which recognises the construct as a significant predictor of behaviour [5]. Extant information security culture literature recognises that in situations of limited time lag between the implied behavioural intention and the actual behaviour, the correlation between the two is reinforced [22]; [5]; [23]. Information security risks normally require prompt decision-making such as in social engineering or opening suspicious links.

According to the findings of this study, subjective norms are a predictor of financial services information security culture. The finding concurs with the prevalent views of extant information security literature which recognises that subjective norms are structured by the way of collective beliefs of a person's significant others [18]. In the financial services setting, the said significant others will be the peers, IT security personnel and management. Information systems quality is proposed to be a determinant of the manner in which subjective norms are invoked by employees [18]. Financial services firms are normally in a position to acquire state-of-the-art information technologies which helps to explain the findings of this study with regards to subjective norms. However, there are also extant literature assertions to the

effect that subjective norms are an unreliable construct in the determination of information security culture [35]; [36]. Nevertheless, the finding in relation to subjective norms in this study seems to be predicated on the proposal of [33] that social influence plays a more important role in low-income countries as they are characterised by a very limited number of individuals with high information security awareness.

The study's finding with regards to espoused values as not being a significant determinant of financial services information security culture goes against the grain. Extant literature on information security culture contends that espoused values are a key determinant of information security culture. Specifically, the Value-Attitude-Behaviour Theory posits that espoused values influence behaviour, though that influence is intermediated by attitude [17].

6.2 Implications

The fact that knowledge, attitude, SN, STA and BI have been proved to be significant predictors of financial services information security culture in Zimbabwe implies that top management in the financial services firms in the country and similar jurisdictions should pay close attention to the implementation of those constructs in their respective firms. Additionally, the fact that social constructs take sway in the global south than they do in the global north [33] implies that financial services top management should promote the cultivation of information security culture through such measures as information security knowledge sharing and general socialisation amongst their employees.

This study recommends that governments in the global south improve the provision of information security culture education to general populace either through formal or vocational education in order to improve the national information security knowledge, awareness and subjective norms. Apart from financial services firms recruiting from the society where the norms attuned to information security are prevalent, the fact that the constructs of this study are socially construed [33] implies that an information security aware society will produce an even more robust information security culture in the financial services firms.

Given that the information security terrain is ever shifting with new threat vectors surfacing on a regular basis, financial services firms should consider conducting frequent information security awareness and information security training and education sessions within their respective firms. These will help improve the knowledge, attitudes and subjective norms of their employees which have all been empirically proven to be antecedents of financial services firms' information security culture by this study.

7 Limitations and conclusions

7.1 Limitations

Though this study collected a large sample, its purpose was to generalise to the concept of financial services information security culture as opposed to the population. The first limitation that the study was subjected to was that it was a cross-sectional survey instead a longitudinal survey. Therefore, there is a chance that the results of the study could have been different had a longitudinal study have been performed. The second limitation of this study was that it was a quantitative study which implies that it missed the opportunity to enquire on the reasons why the respondents provided the responses that they did especially with regards to the SN and EV where the results went against the grain. Qualitative interviews would have been able to shed more light on those findings. Readers are made aware of the potential confounders of those limitations.

Future studies should consider the performance of longitudinal studies to examine the influence of knowledge, attitude, SN, STA, EV and BI on financial services firms' information security culture to further the understanding of those relationships. The performance of qualitative interviews will also help shed light on the rationale of the respondents' choices of responses to the questionnaire.

7.2 Conclusions

This study examined the effect that knowledge, attitude, SN, STA, EV and BI have financial services information security culture. It established that knowledge, attitude, SN, STA and BI are predictors of financial services information security culture while EV was found not to be a predictor of financial services information security culture.

The current study was one of the seminal studies to empirically examine the impact of knowledge, attitude, SN, STA, EV and BI on financial services firms' information security culture. It contributed to methodology and practice guidance on the topical information security subfield of information security culture. This study also serves as a robust springboard for future investigations on the same subject matter though its results need to be confirmed by future results of qualitative or mixed-methods studies.

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Declarations

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